

Assignment 2 – Tapeless ruler

Description:

With the HC-SR04 echo sensor, a C program will be made to continuously measure the distance of an object in front of the sensor, which will update in real time. The echo sensor will be connected to the raspberry pi via breadboard and gpio connections and the program will be written with the PiGPIO C library.

Approach / What I Did:

I began by connecting the HC-SR04 echo sensor and the Raspberry Pi through the breadboard. Since the technical documentation states that the echo sensor's working voltage is 5 volts, I decided to use the 1K ohm resistors for connections from the echo sensor to the pi which are the echo pin and the ground pin. Other than that, the connections were straightforward a one-to-one connection for each pin to the Raspberry Pi.

As for the code, I went with the PiGPIO library since it is well documented and already provides lots of useful functions for working with the GPIO pins as I saw when I worked on assignment 1 traffic light.

Since the program will run in a loop until the user decides to stop, I had to figure out a way to handle user interrupts like ctrl c and ctrl z when the user stops the program. For this, I used the signal.h library, which I found online documentation on how to use my own defined function to handle signals using a struct called sigaction. I handled user interrupt signals by making sure the pigpio library was terminated before exiting the program, so as to terminate the program gracefully.

To record the time that the echo receiver was pulled up, I used two while loops. The first while loop repeatedly recorded the current time as the start time while the echo pin was down but as soon as it pulled up, the last recorded time was ultimately recorded as the real start time. Then the second while loop repeatedly records the current time as the end time as the ECHO pin is pulled up but once it is pulled down through the reception of ultrasonic, the last recorded time is the real end time and those numbers are used for calculations.

Issues and Resolutions:

The first issue I came across was understanding how to record times when the echo pin is pulled up. At first, I thought I had to manually pull it up after starting the trigger pulse, but that did not make sense since that pulling up statement could execute at a different time than when the trigger pulse happens. I later realized that the trigger pin on the sensor automatically pulls the echo pin when it is pulsed.

I realized that times could also be recorded with while loops in order to constantly monitor the state of the echo pin and have better accuracy of when ultrasonic is transmitted and received.

Another issue I had was figuring out how to use the signal.h library to handle interrupt signals during the program. When I first began using the library I simply used the signal() function to handle signals but everytime I compiled my program I would get an error like

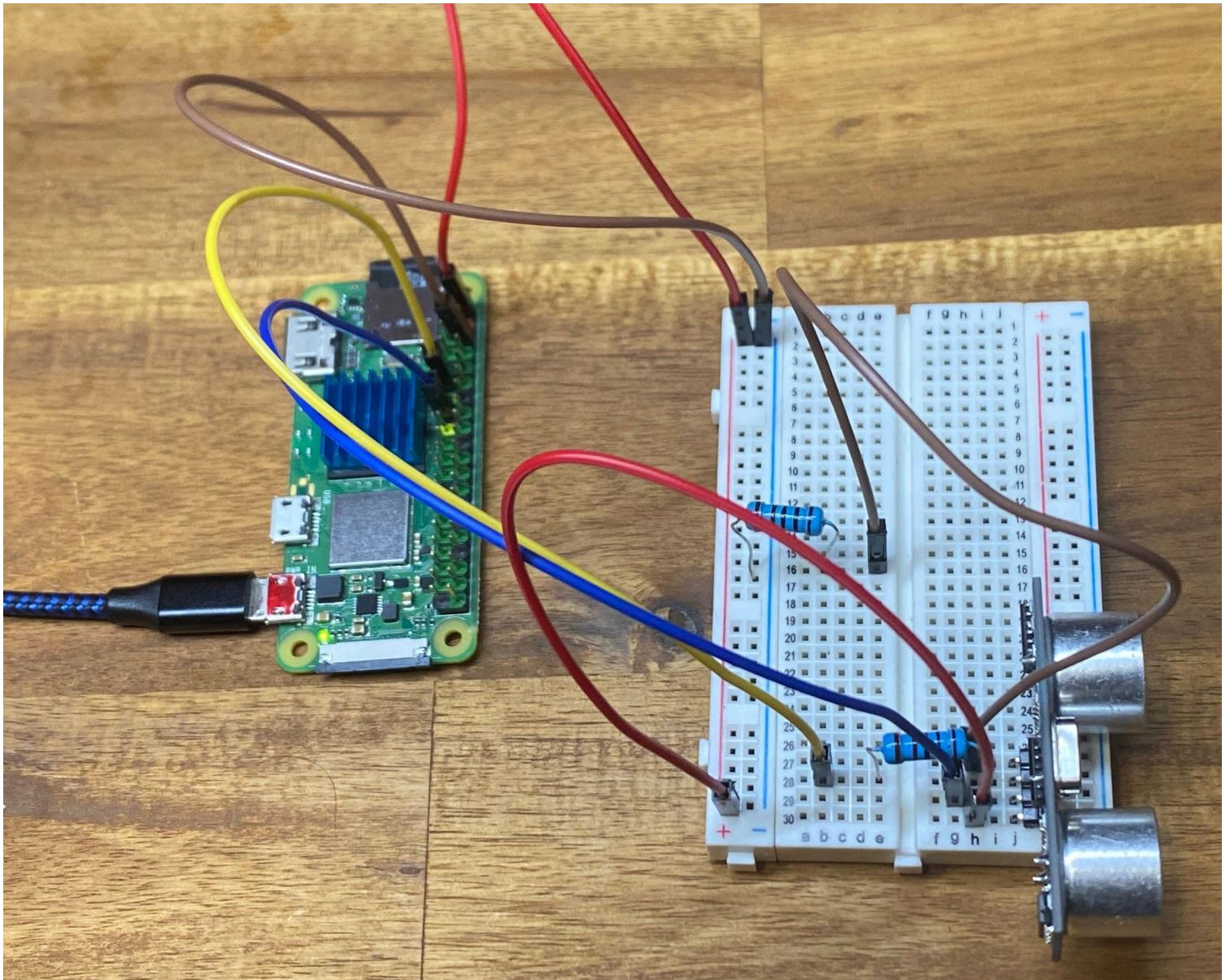
```
/usr/bin/ld: warning: /tmp/cc9ufMqg.o: requires executable stack (because the .note.GNU-stack section is executable)
```

I found online that this was a security issue due to the nature of it involving executing code coming from interrupts. Online resources gave solutions such as adding compiler flags that would ignore the warning but a more verbose usage of the signal.h library made the warning go away by using the sigaction struct.

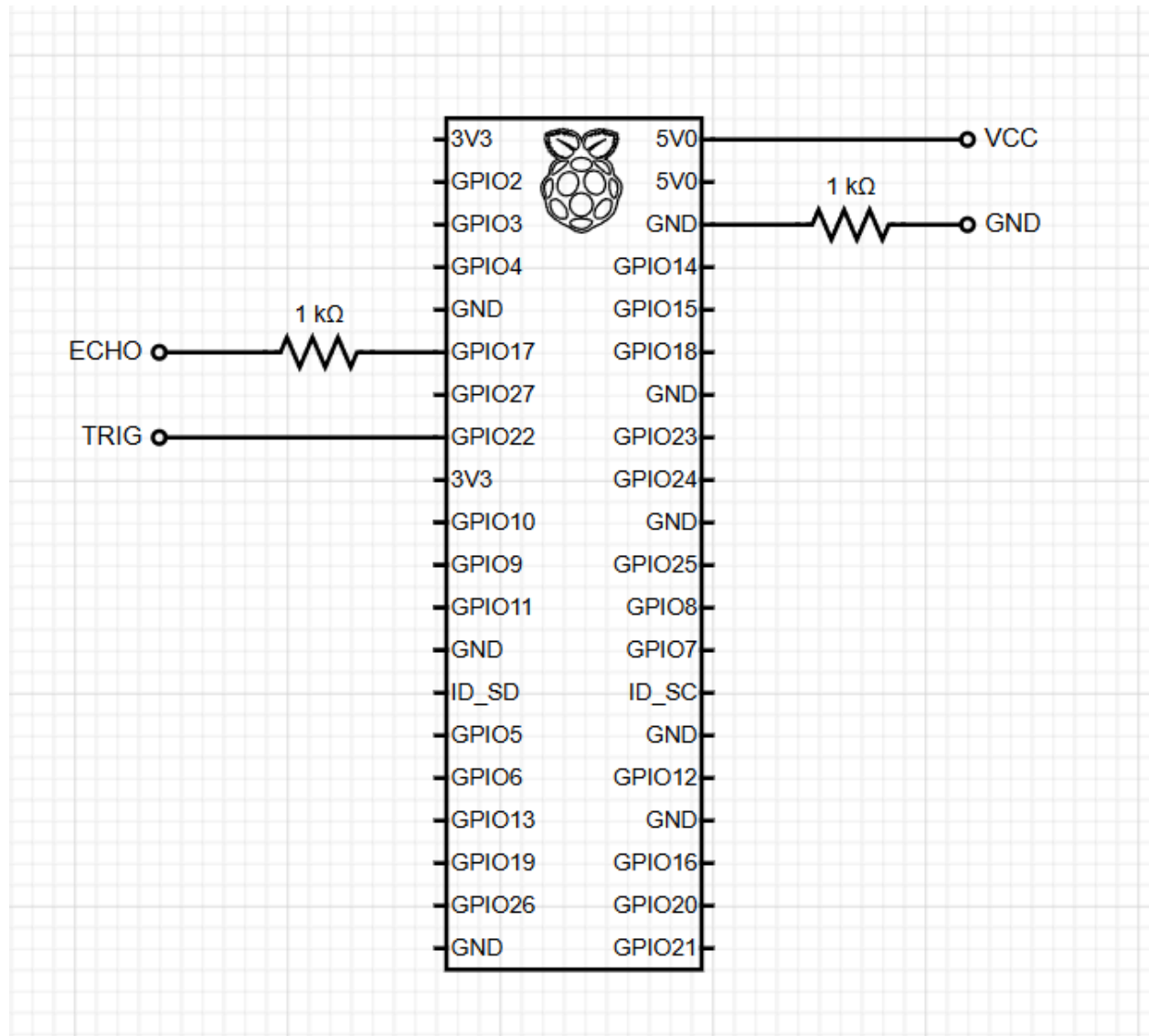
```
//Setup to safely handle interrupt signal and exit program
struct sigaction sa;
sa.sa_handler = handleUsrInterrupt;
sigemptyset(&sa.sa_mask);
sa.sa_flags = 0;
```

```
if(mainInitilize() < 0){
    printf("Object distance: %.2fcm\n", dista
//Handle interrupt signal
sigaction(SIGINT, &sa, NULL);
sigaction(SIGTSTP, &sa, NULL);
```

Photo of completed circuit:



Hardware Diagram:



Analysis:

(If required for the assignment)

Screen shot of compilation:

```
pi42@raspberrypi42:~/CSC615/csc-615-assignment-2-tapeless-ruler-AmgBar $ make clean
rm -f assignment2
pi42@raspberrypi42:~/CSC615/csc-615-assignment-2-tapeless-ruler-AmgBar $ make
gcc -Wall -pthread -o assignment2 assignment2.c -lpigpio -lrt
pi42@raspberrypi42:~/CSC615/csc-615-assignment-2-tapeless-ruler-AmgBar $
```

Screen shot(s) of the execution of the program:

```
PROBLEMS OUTPUT TERMINAL ... bash - csc-615-assignment-2-tapeless-ruler-AmgBar + v [ ] [X]
pi42@raspberrypi42:~/CSC615/csc-615-assignment-2-tapeless-ruler-AmgBar $ make clean
rm -f assignment2
pi42@raspberrypi42:~/CSC615/csc-615-assignment-2-tapeless-ruler-AmgBar $ make
gcc -Wall -pthread -o assignment2 assignment2.c -lpigpio -lrt
pi42@raspberrypi42:~/CSC615/csc-615-assignment-2-tapeless-ruler-AmgBar $ make run
sudo ./assignment2
Object distance: 5.37cm
Object distance: 5.07cm
Object distance: 5.61cm
Object distance: 6.85cm
Object distance: 8.40cm
Object distance: 9.91cm
Object distance: 11.36cm
Object distance: 12.90cm
Object distance: 13.31cm
Object distance: 14.55cm
Object distance: 15.62cm
Object distance: 16.51cm
Object distance: 16.81cm
Object distance: 17.22cm
Object distance: 17.22cm
Object distance: 16.81cm
Object distance: 17.22cm
^C
User interrupt detected...exiting tapeless ruler
pi42@raspberrypi42:~/CSC615/csc-615-assignment-2-tapeless-ruler-AmgBar $
```